

1. Project Title

AI-Based Bus Passenger Demand Prediction and Extra Bus Alert System

Location: Tirupur Bus Stand

Pathway: Pathway A – Continuation Track

Department: Computer Science and Engineering

2. Problem Statement

Passengers at Tirupur Bus Stand wait for scheduled buses while the crowd gradually increases as the bus arrival time approaches. Current crowd information alone cannot indicate how many passengers may arrive before the bus reaches the bus stand.

This uncertainty can lead to overcrowding and difficulty in managing bus capacity.

Problem Definition:

How can we estimate future passenger demand before bus arrival?

3. Proposed Solution

The proposed system is an **AI-Based Bus Passenger Demand Prediction and Extra Bus Alert System**.

The system uses the current waiting passengers, historical average, bus capacity, scheduled arrival information, and remaining time to estimate the expected passenger demand before the bus arrives.

The predicted demand is then compared with the bus capacity.

If predicted demand exceeds capacity:

HIGH CROWD EXPECTED

EXTRA BUS RECOMMENDED

Otherwise:

NORMAL CROWD EXPECTED

EXTRA BUS NOT REQUIRED

4. Objective

- Predict future passenger demand before bus arrival.
- Compare predicted demand with bus capacity.

- Identify possible overcrowding in advance.
 - Provide an early extra-bus recommendation.
 - Develop a prototype that can later be integrated with a real ML model and live data.
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5. System Workflow

Input Data → Demand Prediction → Capacity Comparison → Crowd Status → Extra Bus Alert

Inputs:

- Bus route/name
- Scheduled arrival time
- Bus capacity
- Current waiting passengers
- Historical average passengers
- Minutes remaining

Output:

- Predicted passenger demand
 - Capacity comparison
 - Crowd status
 - Extra bus recommendation
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6. Review 1 – Completed Work

The following foundational components have been completed:

- Problem identification
- Solution design
- Responsive web dashboard
- Passenger input module
- Demand prediction prototype
- Bus capacity comparison

- Crowd status identification
 - Extra bus recommendation
 - High-crowd test scenario
 - Normal-crowd test scenario
 - GitHub repository
 - Project documentation
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7. Technologies Used

- **HTML** – Webpage structure
 - **CSS** – User interface and responsive design
 - **JavaScript** – Prediction calculation and system logic
 - **GitHub** – Version control and project repository
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8. Current Prediction Method

For the Review 1 prototype, a simple rule-based calculation is used in JavaScript.

The calculation estimates new passengers expected before bus arrival and adds them to the current waiting passengers.

Important: This is a prototype calculation and **not a trained Machine Learning model**.

The prediction logic is kept separately so that a real ML model can be integrated in a future review.

9. Working Evidence

Test Case 1 – High Crowd

Input:

- Current waiting passengers: 42
- Bus capacity: 50
- Historical average: 60
- Minutes remaining: 15

Predicted demand: 72

Result:

HIGH CROWD EXPECTED
EXTRA BUS RECOMMENDED

Quick Demo Scenarios:
Click to test system conditions immediately

Scenario 1: High Surge (Extra Bus Needed)

Scenario 2: Normal Flow (Capacity Sufficient)

Reset Defaults

Live Terminal & Schedule Parameters
Enter real-time crowd and timetable inputs for the upcoming scheduled trip

Bus Route / Name

Route 20A - Tirupur Old Bus Stand to Avinashi

Scheduled Bus Arrival Time

16:45

Bus Capacity (Passengers)

50

seats

Current Waiting Passengers

42

people

Historical Average Passengers

60

people

Observed headcount at the boarding bay

Minutes Remaining Before Bus Arrival

15

mins

Estimated time until the bus arrives at the platform

Predict Demand & Evaluate

Prediction & Extra Bus Evaluation
Real-time demand projection and fleet dispatch alert

 **HIGH CROWD EXPECTED**
EXTRA BUS RECOMMENDED

PREDICTED PASSENGER DEMAND

72 pax
Includes ~30 incoming

SCHEDULED BUS CAPACITY

50 seats
Vehicle baseline limit

DEMAND VS CAPACITY DIFF

+22 pax
22 passengers above capacity

CROWD STATUS

HIGH
144% capacity utilization

DEMAND CALCULATION BREAKDOWN

Current waiting passengers:

42 passengers

Projected new arrivals before bus arrives:

+30 passengers

Estimated arrival rate:

2.0 passengers/min

Total predicted passengers at arrival:

72 passengers (Exceeds Capacity)

Test Case 2 – Normal Crowd

Input:

- Current waiting passengers: 20
- Bus capacity: 55
- Historical average: 35
- Minutes remaining: 10

Predicted demand: 32

Result:

NORMAL CROWD EXPECTED
EXTRA BUS NOT REQUIRED

Quick Demo Scenarios:
 Click to test system conditions immediately

Scenario 1: High Surge (Extra Bus Needed)

Scenario 2: Normal Flow (Capacity Sufficient)

Reset Defaults

Live Terminal & Schedule Parameters
 Enter real-time crowd and timetable inputs for the upcoming scheduled trip

Bus Route / Name
 Route 12B - Tirupur New Bus Stand to Kangeyam (Mid-day)

Scheduled Bus Arrival Time
 14:15

Bus Capacity (Passengers)
 55 seats

Current Waiting Passengers
 20 people

Historical Average Passengers
 35 people

Minutes Remaining Before Bus Arrival
 10 mins

Predict Demand & Evaluate

Prediction & Extra Bus Evaluation
 Real-time demand projection and fleet dispatch alert

NORMAL CROWD EXPECTED
 EXTRA BUS NOT REQUIRED

PREDICTED PASSENGER DEMAND
 32 pax
 Includes ~12 incoming

SCHEDULED BUS CAPACITY
 55 seats
 Vehicle baseline limit

DEMAND VS CAPACITY DIFF
 -23 pax
 23 spare seats remaining

CROWD STATUS
 NORMAL
 58% capacity utilization

DEMAND CALCULATION BREAKDOWN

Current waiting passengers:	20 passengers
Projected new arrivals before bus arrives:	+12 passengers
Estimated arrival rate:	1.2 passengers/min
Total predicted passengers at arrival:	32 passengers (Within Capacity)

10. Current Limitations

The current Review 1 prototype has the following limitations:

- Passenger values are entered manually.
- A real historical dataset is not yet integrated.
- A trained ML model is not yet implemented.
- CCTV/live passenger detection is not yet integrated.
- Automatic notification to transport authorities is not yet implemented.

11. Pending Work / Next Review

The following features will be developed in the next stage:

- Collection of real historical passenger data
- Creation of a proper historical dataset
- Training and integration of an ML prediction model
- Real-time passenger detection
- CCTV/computer vision integration
- Improved prediction accuracy
- Automatic alert/notification system

- Further testing using real-world data
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12. GitHub Repository

Repository:

<https://github.com/tkeerthana4408-tech/AI-bus-demand-prediction.git>

The repository contains the current project source files and documentation.

13. Conclusion

The Review 1 prototype demonstrates the basic working flow of the proposed system: **enter passenger and bus information → predict future demand → compare with capacity → generate an early crowd alert.**

The current prototype establishes the foundation for integrating real historical data and a trained ML model in the next stage.